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DOISR Movie Ontology Group 5 Project

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Introduction

This project is based mainly on the dataset "U.S. movies with gender-disambiguated actors, directors and producers", available on Figshare. In addition to this dataset, we also analyzed and used information from several movie-related datasets to better understand the domain and improve the ontology design. These additional datasets included the "9000 Movies Dataset", the "IMDb 5000 Movie Dataset", and the "IMDb Movie Reviews Genres Description and Emotions" dataset, all available on Kaggle. With this data, we needed to create an OWL ontology that follows FAIR guidelines to make the ontology interoperable and reusable.
To do so we have analyzed the data, found requirements, created an ontology, analysed said ontology, and published it. In this document we will further explain how we did this.

Development

The ontological requirements identified

Below is the requirements we have identified:

Identifier	Competency Question / Natural language sentence (fact)	Answer (For Competency Questions)	Status (Proposed, Accepted, Rejected, Deprecated)	Comments (optional)
ActorR1	what is the name of the actor?	The value of the "name" attribute (String)	Proposed	
ActorR2	What movies has the actor acted in ?	List of movie IDs from "movies_list" (Array of strings)	Proposed	List of movies where the actor participated
ActorR3	what is the gender of the actor?	The value of the "gender" attribute (String)	Proposed	
DirectorR1	what is the name of the Director?	The value of the "name" attribute (String)	Proposed	
DirectorR2	what is the gender of the Director?	The value of the "gender" attribute (String)	Proposed	
DirectorR3	which movies has the director participated in?	List of "movie_id" where the person is "type": "director" inside "movies_list" array	Proposed	
DirectorR4	What is the first movie of the director?	The value of "first_movie" (Integer)	Proposed	
DirectorR5	What is the last movie of the director?	The value of "last_movie" (Integer)	Proposed	
MovieR1	What is the title of the movie?	The value of the "title" attribute (String)	Proposed	
MovieR2	How popular is the movie?	The value of "popularity" (integer)	Proposed	based on the number of views per day, votes per day, number of users marked

				it as "favorite" and "watchlist"
MovieR3	Total votes received from the viewers.	The value of “Vote_Count “ (Integer)	Proposed	
MovieR4	Average rating based on vote count and the number of viewers out of 10.	the value is a number integer from 1 to 10 of ” Vote_Average”	Proposed	
MovieR5	Original language of the movies. Dubbed version is not considered to be original language.	The value of “Original_Language” (String)	Proposed	
MovieR6	Url of the movie poster.	The value of “Poster_Url” (String)	Proposed	
MovieR7	What are the genres of the movie?	The values of the "genre" atrribute (Array of strings)	Proposed	
MovieR8	Who are the directors of the movie?	The value of the attribute "name" in the field "director" (String)	Proposed	
MovieR9	Who are the actos participated in the movie?	The values of the attribute "name" in the field of "all_actors" (Array of strings)	Proposed	
MovieR10	Date when the movie was released.	The value of the atrribute "year" (Integer)	Proposed	
MovieR11	Written analyses of the films that share thoughts and observations.	The value of “reviews” (String)	Proposed	
MovieR12	Each description expresses an emotional tone.	the value of “Emotions” (String)	Proposed	sadness,anticipation,joy,fear,optimism
MovieR13	Brief summary of the movie.	The value of “Overview” (String)	Proposed	
MovieR14	The percentage of female actors in the movie	The value of the "gender_percent" atrribute (Integer)	Proposed	
MovieR15	What was the budget of the movie?	The value of the "adjusted_budget" attribute (Integer)	Proposed	
MovieR16	What is the movie duration?	The value of “duration” in minutes (Integer)	Proposed	
ProducerR1	What is the first movie of the Producer?	The value of the "first_movie" attribute (Integer)	Proposed	
ProducerR2	which movies has the producer participated in?	List of "movie_id" inside "movies_list" array	Proposed	
ProducerR3	What is the last movie of the Producer?	The value of the "last_movie" attribute (Integer)	Proposed	

ProducerR4	what is the gender of the producer?	The value of the "gender" attribute (String)	Proposed

The main design decisions made during the development of ontologies

One of the biggest things we learned during the project was how different ontology engineering is from designing a traditional database. In our first ontology version, some concepts such as actors and directors were incorrectly modeled as strings instead of semantic entities. After reviewing the ontology and receiving feedback, we redesigned those concepts as classes connected through object properties.

We also removed generic relationships such as "has" and "is" because they were too ambiguous and did not clearly describe the semantic meaning of the relationships. Instead, we replaced them with more meaningful properties such as hasActor, hasDirector, hasProducer, belongsToGenre, and hasOriginalLanguage.

Another important design decision was separating object properties and datatype properties correctly. In the initial version we mixed both types of properties, but later we reorganized the ontology so that object properties only connected classes and datatype properties only connected literal values.

We also decided to use xsd:date for dates instead of creating a separate Date class and represented movie duration using a datatype property called durationMinutes. Controlled vocabularies were also used for genres, languages, and gender categories to make the ontology more consistent and semantically clear.

The reused ontological resources and the explanation of the reuse process followed

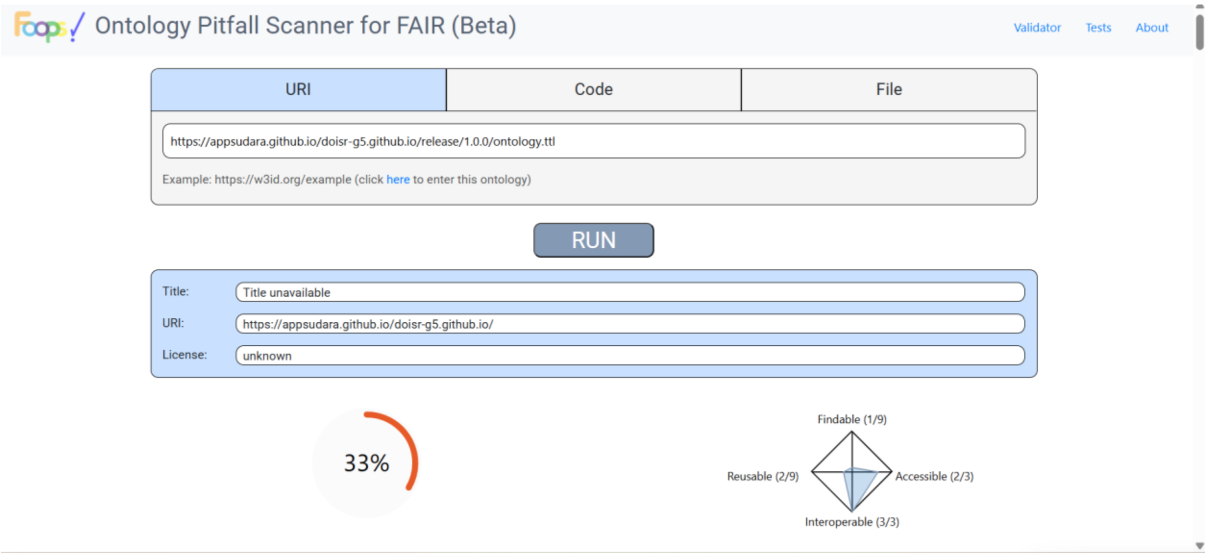
Our reuse process mainly consisted of searching for ontologies related to the movie domain, analyzing whether their concepts matched our needs, and deciding which ideas or semantic structures could be reused or adapted in our ontology. Instead of importing complete ontologies, we mainly reused concepts, naming conventions, and modeling approaches.

FOAF helped us understand how people are commonly represented in ontologies, while schema.org was useful for comparing movie-related concepts and metadata. SKOS was especially helpful for understanding how controlled vocabularies such as genres and languages could be modelled semantically.

The evaluation techniques and tools used as well as the design decisions or changes made on the ontology according to the evaluation results

We created SHACL shapes to validate our data according to the ontology constraints. These shapes allowed us to validate datatypes, cardinality restrictions, string lengths, numeric ranges, and class relationships. In order to test the validation process, we created both valid and invalid examples that intentionally violated some constraints. This helped us better understand how validation works and allowed us to identify issues such as missing required properties, invalid datatypes, incorrect relationships, and negative numeric values.

The documentation system chosen and the license that has been declared for the ontologies and report on the ontology FAIRness level



To evaluate the FAIRness level of the ontology, we used FOOPS!. We submitted the ontology URI to the tool to analyze how well the ontology follows the FAIR principles. The ontology obtained a FAIRness score of 33%. According to the FOOPS! evaluation, the ontology performed well in terms of interoperability, obtaining the maximum score in that category. Accessibility also obtained positive results since the ontology could be accessed through its public URI. However, the evaluation showed that there is still room for improvement regarding findability and reusability. Even though the FAIRness score was not very high, the evaluation helped us better understand the importance of metadata, documentation, persistent identifiers, and ontology publication practices when creating semantic resources. The FOOPS! analysis was useful for identifying improvements that could be made in future versions of the ontology.

The additional tools used

During the development of this project, we have used ChatGPT as a support tool. Since ontology engineering and semantic web technologies were relatively new topics for us, we used it mainly to ask questions, better understand certain concepts, and receive guidance on how to implement different parts of the ontology and SHACL validation. However, ChatGPT was only used as an aid during the learning and development process. All the generated content, ontology structures, and validation rules were carefully reviewed, modified when necessary, and understood by us before being included in the final project. The tool helped speed up the learning process and resolve doubts, but the final design decisions and implementation work were done by our group.

Any other comments that the student considers necessary

One of the main difficulties during our project was learning to think semantically instead of modeling everything like a traditional database. At the beginning, several concepts were represented incorrectly because the ontology was treated too much like a relational schema. After multiple iterations and feedback, the ontology became much cleaner and more aligned with OWL and semantic web best practices.

Conclusions

Throughout this project we learned many things about ontology engineering and semantic web technologies. One of the most important parts was understanding how to move from thinking about data as a traditional database to thinking about it semantically using OWL and SHACL. We also learned what FAIR principles are and why they are important when creating reusable and interoperable semantic resources. During the project, we improved our ability to analyze datasets and transform them into structured knowledge that can be validated and reused in a more meaningful way. Another important lesson was understanding the difference between classes, object properties, and datatype properties, as well as learning how ontology design decisions can greatly affect the quality and clarity of the final ontology. Using tools such as Chowlk, Protégé, FOOPS, and SHACL validation helped us better understand the ontology development process and how semantic resources are evaluated and improved iteratively. Overall, the project gave us practical experience with ontology development and helped us understand how semantic technologies and FAIR principles can be applied to real-world datasets in a useful and structured way.